

Unit 6: Multivariable Calculus Foundations

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 6 at a Glance: Core Sub-Topics

Differential equations in more than one variable need the language of multivariable calculus. This unit builds that bridge: we learn to differentiate one variable at a time, to package those partials into a gradient, to chain through intermediate variables, to locate and classify extrema, and to read the geometry of level curves — the same geometry that decides when a differential $M dx + N dy$ is *exact*.

- **Partial Derivatives** — differentiate in one variable while holding the others constant; f_x, f_y .
- **Higher-Order and Mixed Partial** — Clairaut's theorem: $f_{xy} = f_{yx}$ when the second partials are continuous.
- **Gradient and Directional Derivatives** — ∇f collects the partials; $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ measures change along a unit direction.
- **The Multivariable Chain Rule** — sum one product term per path from the output down to the independent variable.
- **Critical Points and Optimization** — set $\nabla f = \mathbf{0}$, then classify with the second-derivative (Hessian) test.
- **Level Curves and Exactness** — $f(x, y) = c$, ∇f normal to the curve, and the exact-differential test $\partial_y M = \partial_x N$.

Part I — Condensed Cheat Sheet

6.1 Partial Derivatives

A partial derivative differentiates a multivariable function in *one* variable while treating every other variable as a constant symbol. It measures the instantaneous rate of change of f along a single coordinate axis.

Definition of the Partial Derivatives

$$f_x = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad f_y = \frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}.$$

- To compute f_x , **hold y constant** and differentiate in x with the ordinary rules; any pure- y term is a constant and dies.
- Holding a variable *constant* is not the same as *setting it to zero*: the symbol stays in the answer.

Worked Example 6.1: First partials of $f = x^2y + y^3$

Treat y as constant for f_x , then x as constant for f_y :

$$f_x = 2xy + 0 = 2xy, \quad f_y = x^2 + 3y^2.$$

The y^3 term vanishes under ∂_x , while the x^2 term contributes its coefficient x^2 under ∂_y .

6.2 Higher-Order and Mixed Partial

Each first partial can itself be differentiated again, giving four second-order partials: two *pure* (f_{xx}, f_{yy}) and two *mixed* (f_{xy}, f_{yx}). Clairaut's theorem says the mixed pair agrees whenever the function is smooth enough.

Clairaut's Theorem on Mixed Partial

If the second partial derivatives of f are **continuous** near a point, then the mixed partials are equal there:

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = f_{yx}.$$

The order of differentiation does not matter — a function of two variables has **four** second partials but only **three** distinct values.

Worked Example 6.2: Mixed partials of $f = x^2y^3$

Differentiate in x first, then in y — and the reverse:

$$f_x = 2xy^3 \Rightarrow f_{xy} = 6xy^2, \quad f_y = 3x^2y^2 \Rightarrow f_{yx} = 6xy^2.$$

Both routes land on $6xy^2$, exactly as Clairaut guarantees.

6.3 The Gradient and Directional Derivatives

The gradient packages the partials into a single vector that points in the direction of steepest increase. Dotting it with a *unit* direction gives the rate of change along that direction — the directional derivative.

Gradient, Directional Derivative, and Steepest Ascent

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right), \quad D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} \quad (\mathbf{u} \text{ a unit vector}).$$

- The **maximum** rate of increase at a point is $|\nabla f|$, attained in the direction $\mathbf{u} = \nabla f / |\nabla f|$.
- ∇f is **perpendicular** to the level curve through the point.

Common Pitfall: The Direction Must Be a Unit Vector

$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ is only valid when $\mathbf{u}$ is **normalized**. Given a raw direction $\mathbf{v}$, divide first: $\mathbf{u} = \mathbf{v}/|\mathbf{v}|$. Skipping the normalization rescales the answer by $|\mathbf{v}|$ and silently corrupts every directional derivative and max-rate computation that follows.

Worked Example 6.3: Directional derivative of $f = x^2 + xy + y^2$

At $(1, 2)$, $\nabla f = (2x + y, x + 2y) = (4, 5)$. Along $\mathbf{v} = (3, 4)$, normalize to $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$:

$$D_{\mathbf{u}}f = (4, 5) \cdot \left(\frac{3}{5}, \frac{4}{5} \right) = \frac{12}{5} + \frac{20}{5} = \frac{32}{5} = 6.4.$$

The steepest ascent there points along $(4, 5)$ with rate $|\nabla f| = \sqrt{41} \approx 6.40$.

6.4 The Multivariable Chain Rule

When the inputs of f are themselves functions, differentiate by summing one product term for every path from the output down to the variable you are differentiating with respect to. A small *tree diagram* bookkeeps the paths.

Chain Rule Trees: Counting the Paths

One independent variable $z = f(x(t), y(t))$ — two paths to t :

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = \nabla f \cdot (x', y').$$

Two independent variables $z = f(x(s, t), y(s, t))$ — two paths to each:

$$\frac{\partial z}{\partial s} = f_x x_s + f_y y_s, \quad \frac{\partial z}{\partial t} = f_x x_t + f_y y_t.$$

1. Draw a branch from z to each intermediate variable (x, y) .
2. Extend a branch from each intermediate variable to the target (t) .
3. **Multiply** along each branch, then **add** the branches.

Worked Example 6.4: Chain rule for $z = x^2y$, $x = t^2$, $y = t^3$

With $z_x = 2xy$, $z_y = x^2$, $x' = 2t$, $y' = 3t^2$:

$$\frac{dz}{dt} = (2xy)(2t) + (x^2)(3t^2) = (2t^5)(2t) + (t^4)(3t^2) = 4t^6 + 3t^6 = 7t^6.$$

The direct substitution $z = t^4 \cdot t^3 = t^7$ confirms $dz/dt = 7t^6$.

6.5 Critical Points and Optimization

Interior extrema of a smooth $f(x, y)$ occur where the gradient vanishes. The *second-derivative test* then uses the sign of the Hessian discriminant to sort each critical point into a minimum, maximum, or saddle.

Locating and Classifying Critical Points

1. **Solve** $\nabla f = \mathbf{0}$, i.e. $f_x = 0$ and $f_y = 0$ simultaneously, to find all critical points.
2. **Form** the discriminant $D = f_{xx}f_{yy} - (f_{xy})^2$ at each point.
3. **Classify** with D and f_{xx} (Step in the box below).

The Second-Derivative (Hessian) Test

$$D = f_{xx}f_{yy} - (f_{xy})^2.$$

- $D > 0$, $f_{xx} > 0$: **local minimum**.
- $D > 0$, $f_{xx} < 0$: **local maximum**.
- $D < 0$: **saddle point** (no extremum).
- $D = 0$: the test is **inconclusive**.

Common Pitfall: Reading the Sign Boundary Wrong

The sign of f_{xx} only decides min vs. max *after* $D > 0$ confirms an extremum. When $D < 0$ the point is a **saddle** regardless of f_{xx} , and when $D = 0$ the test says *nothing* — do not declare a saddle by default. A single mishandled sign flips a minimum into a maximum.

6.6 Level Curves, Gradients, and Exactness

Level curves $f(x, y) = c$ slice a surface into contour lines; the gradient cuts across them at right angles. This same gradient structure is exactly what makes a first-order differential *exact* — the payoff that carries into Unit 7.

Level Curves and the Exact-Differential Test

A **level curve** is the set $\{(x, y) : f(x, y) = c\}$; ∇f is normal to it. A differential $M(x, y) dx + N(x, y) dy$ is **exact** on a simply connected region exactly when

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

in which case it equals df for a potential f with $f_x = M$, $f_y = N$ — and the field $\mathbf{F} = (M, N) = \nabla f$ is **conservative**.

Common Pitfall: Path Dependence and the Wrong Cross Partial

The exactness/conservative test compares the *cross* partials $\partial_y M$ and $\partial_x N$ — not $\partial_x M$ and $\partial_y N$. When they differ the field is path-dependent: $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends on the route, so there is *no* single potential to read off.

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from first and mixed partial derivatives, through the gradient and directional derivatives, into the multivariable chain rule, and close with a full optimization sweep — finding and classifying critical points with the second-derivative test. Solutions follow in Part III.

1. For $f(x, y) = x^2y + y^3$, compute both first partials f_x and f_y .
2. For $f(x, y) = x^2y^3$, compute the mixed partials f_{xy} and f_{yx} and confirm Clairaut's theorem.
3. For $f(x, y) = x^2 + xy + y^2$, find the gradient ∇f and evaluate it at the point $(1, 2)$.
4. Using the gradient from Problem 3, compute the directional derivative of f at $(1, 2)$ in the direction of $\mathbf{v} = (3, 4)$. Remember to normalize.
5. At the same point $(1, 2)$, state the direction of steepest ascent of $f = x^2 + xy + y^2$ and the maximum rate of increase there.
6. For $z = x^2 + y^2$ with $x = \cos t$ and $y = \sin t$, use the chain rule to find $\frac{dz}{dt}$, and explain the result geometrically.
7. For $z = x^2y$ with $x = s + t$ and $y = st$, use the chain-rule tree to find $\frac{\partial z}{\partial s}$.
8. Find and classify the single critical point of $f(x, y) = x^2 + y^2 - 4x - 6y + 13$, and give the extreme value.
9. Find and classify the critical point of $f(x, y) = x^2 - y^2$ using the second-derivative test.
10. **(Capstone)** Find *all* critical points of $f(x, y) = x^3 - 3x + y^2$ and classify each with the second-derivative test, reporting any extreme values.

Part III — Complete Solutions

Solution 1. Hold y constant for f_x , then x constant for f_y . The pure- y term y^3 vanishes under ∂_x :

$$f_x = 2xy, \quad f_y = x^2 + 3y^2.$$

Solution 2. Differentiate in x first, then y — and in the reverse order:

$$f_x = 2xy^3 \Rightarrow f_{xy} = 6xy^2, \quad f_y = 3x^2y^2 \Rightarrow f_{yx} = 6xy^2.$$

Both equal $6xy^2$, so $f_{xy} = f_{yx}$ — Clairaut's theorem holds because the second partials are continuous everywhere.

Solution 3. Differentiate componentwise:

$$\nabla f = (2x + y, x + 2y), \quad \nabla f(1, 2) = (2 + 2, 1 + 4) = (4, 5).$$

Solution 4. The raw direction $\mathbf{v} = (3, 4)$ has $|\mathbf{v}| = \sqrt{9 + 16} = 5$, so the unit direction is $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$. Then

$$D_{\mathbf{u}}f = \nabla f(1, 2) \cdot \mathbf{u} = (4, 5) \cdot (\frac{3}{5}, \frac{4}{5}) = \frac{12}{5} + \frac{20}{5} = \frac{32}{5} = 6.4.$$

Solution 5. Steepest ascent points along the gradient itself, $\nabla f(1, 2) = (4, 5)$, i.e. the unit direction $(4, 5)/\sqrt{41}$. The maximum rate of increase is the gradient's magnitude:

$$|\nabla f(1, 2)| = \sqrt{4^2 + 5^2} = \sqrt{41} \approx 6.40.$$

Solution 6. With $z_x = 2x$, $z_y = 2y$, $x' = -\sin t$, $y' = \cos t$:

$$\frac{dz}{dt} = (2x)(-\sin t) + (2y)(\cos t) = 2\cos t(-\sin t) + 2\sin t(\cos t) = 0.$$

Geometrically the path $(\cos t, \sin t)$ rides the unit circle, where $z = x^2 + y^2 = 1$ is constant — so its rate of change is zero, and the velocity is tangent to the level curve, perpendicular to ∇z .

Solution 7. The tree has two branches to s : $\frac{\partial z}{\partial s} = z_x x_s + z_y y_s$ with $z_x = 2xy$, $z_y = x^2$, $x_s = 1$, $y_s = t$:

$$\frac{\partial z}{\partial s} = (2xy)(1) + (x^2)(t) = 2(s+t)(st) + (s+t)^2 t.$$

Factoring $t(s+t)$ gives

$$\frac{\partial z}{\partial s} = t(s+t)[2s + (s+t)] = t(s+t)(3s+t).$$

Solution 8. Set the gradient to zero:

$$f_x = 2x - 4 = 0 \Rightarrow x = 2, \quad f_y = 2y - 6 = 0 \Rightarrow y = 3,$$

so the only critical point is $(2, 3)$. The second partials are $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$, giving

$$D = f_{xx}f_{yy} - (f_{xy})^2 = (2)(2) - 0 = 4 > 0, \quad f_{xx} = 2 > 0,$$

a **local minimum**. The value is $f(2, 3) = 4 + 9 - 8 - 18 + 13 = 0$.

Solution 9. Here $f_x = 2x = 0$ and $f_y = -2y = 0$ give the lone critical point $(0, 0)$. With $f_{xx} = 2$, $f_{yy} = -2$, $f_{xy} = 0$:

$$D = (2)(-2) - 0 = -4 < 0,$$

so $(0, 0)$ is a **saddle point** — f rises along x and falls along y , and the sign of f_{xx} is irrelevant once $D < 0$.

Solution (Capstone). Set both partials to zero:

$$f_x = 3x^2 - 3 = 0 \Rightarrow x = \pm 1, \quad f_y = 2y = 0 \Rightarrow y = 0,$$

giving critical points $(1, 0)$ and $(-1, 0)$. The second partials are $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$, so $D = 12x$.

$$(1, 0) : D = 12 > 0, \quad f_{xx} = 6 > 0 \Rightarrow \text{local minimum}, \quad f(1, 0) = 1 - 3 = -2.$$

$$(-1, 0) : D = -12 < 0 \Rightarrow \text{saddle point}.$$

So f has a local minimum value -2 at $(1, 0)$ and a saddle at $(-1, 0)$; the cubic term means there is no global extremum.

Unit 6 Key Takeaway

The workflow is uniform: **partials** build the **gradient** ∇f ; the gradient drives directional derivatives $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ (always on a *unit* $\mathbf{u}$) and points along steepest ascent $|\nabla f|$; the **chain rule** sums one term per path; and **optimization** solves $\nabla f = \mathbf{0}$ then sorts each point with $D = f_{xx}f_{yy} - f_{xy}^2$ — $D > 0$ an extremum (sign of f_{xx} decides min vs. max), $D < 0$ a saddle.